

Natural Language Processing - Quiz 1

25 Marks

Name: _____

Roll no.: _____

Date: 30.01.2026

Instructions: Write all answers on the back-side of the page. Best of luck!

1 MCQ

Multiple options can be correct; all correct options need to be chosen.

Q1: Why are lexical and syntactic processing insufficient for advanced NLP applications such as chatbots and machine translation? [1]

- (a) They increase computational complexity
- (b) They cannot handle large vocabularies
- (c) They require annotated datasets
- (d) They fail to capture the underlying meaning and intent of text

Q2: Given the string "Look!", which of the following patterns will match the character "l"? [1]

- (a) \w
- (b) \d
- (c) \W
- (d) \s

Q3: Which of the following patterns will match the string "Daiyu" completely? [1]

- (a) \D+
- (b) \w+
- (c) [0-9] +
- (d) \S+

Q4: Which of the following statements about WordPiece tokenization is TRUE? [1]

- (a) It merges the most frequent character pairs
- (b) It selects subwords to maximize training data likelihood
- (c) It optimizes likelihood under a unigram language model
- (d) It always produces character-level tokens

Q5: Which technique explicitly models morphology by mapping an input word to its stem and affixes using state transitions? [1]

- (a) Finite State Transducers (FSTs)
- (b) Radix trees
- (c) Neural language models

(d) N-gram models

Q6: Which of the following is a major limitation of N-gram language models? [1]

- (a) They cannot compute joint probabilities
- (b) They require labeled training data
- (c) They cannot be trained on large datasets
- (d) They cannot capture long-distance dependencies

Q7: Which techniques are used in N-gram models to handle unseen word sequences? [1]

- (a) Dropout and attention
- (b) Interpolation and backoff
- (c) Stemming and lemmatization
- (d) Parsing and chunking

Q8: The Maximum Likelihood Estimate (MLE) for a bigram probability is: [1]

- (a) $P(w_i | w_{i-1}) = \frac{1}{V}$
- (b) $\frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$
- (c) $\frac{C(w_i)}{N}$
- (d) $\frac{C(w_i, w_{i+1})}{C(w_i)}$

Q9: Which technique assigns small probabilities to unseen N-grams? [1]

- (a) Stemming
- (b) Tokenization
- (c) Smoothing
- (d) Parsing

Q10: Interpolation differs from backoff because interpolation: [1]

- (a) Uses only lower-order models
- (b) Discards higher-order probabilities
- (c) Combines multiple N-gram models simultaneously
- (d) Assigns zero probability to unseen events

2 Fill in the Blanks

(Q11) The tokenization algorithm used by BERT is _____. [1]

- (Q12) If the length of string ‘X’ is ‘m’ and string ‘Y’ is ‘n’ then the time complexity of computing edit distance using dynamic programming is [1]
- (Q13) The most commonly used intrinsic evaluation metric for language models is [1]
- (Q14) For a fixed test corpus, a language model with lower perplexity assigns probability to the test data. [1]
- (Q15) Backoff language models redistribute probability mass from order N-grams to order N-grams. [1]

Answers

Q1: d

Q2: c

Q3: a, b, d

Q4: b

Q5: a

Q6: d

Q7: b

Q8: b

Q9: c

Q10: c

Q11: WordPiece

Q12: $O(nm)$

Q13: Perplexity

Q14: Higher

Q15: Higher, lower

3 Numerical

- (a) Show the edit distance between “kitten” and “sitting” assuming insertion, deletion and substitution cost is 1. Fill up the table (format of the table given below). Suggest one possible alignment along with the operation applied at each position (i for insertion/ d for deletion/ s for substitution) for achieving optimized edit distance. For showing alignment, write “kitten” on top and place “sitting” at the bottom, and the starting letter of “sitting” should be placed properly to show alignment.

[8+2=10 Marks]

n	6	6	5	4	3	3	2	3
e	5	5	4	3	2	2	3	4
t	4	4	3	2	1	2	3	4
t	3	3	2	1	2	3	4	5
i	2	2	1	2	3	4	5	6
k	1	1	2	3	4	5	6	7
	0	1	2	3	4	5	6	7
		s	i	t	t	i	n	g